

Aravind Gudikandula

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SUMMARY

AI/ML Engineer building production-grade systems across LLMs, RAG, agentic workflows, computer vision, and ML inference. Experience deploying scalable AI services, optimizing systems handling **1M+ daily requests**, and developing end-to-end AI applications across retrieval, evaluation, APIs, CI/CD, monitoring, and production deployment.

EXPERIENCE

- Saayam Foundation** Remote
AI Engineer Feb 2026 – Present
 - Architected agentic AI workflows for **3+ use cases** using **LangGraph**, **LangChain**, **CrewAI**, and **Model Context Protocol (MCP)**, implementing tool calling, persistent state, human-in-the-loop controls, and auditable execution for multi-step workflows.
 - Built hybrid RAG and GraphRAG pipelines across structured and unstructured data using **FAISS**, **Pinecone**, and **OpenSearch**, combining semantic retrieval, hybrid search, and multi-hop reasoning to improve retrieval quality.
 - Improved LLM response quality by **20%** using **LoRA/PEFT** fine-tuning, prompt optimization, and domain-specific training data; implemented **RAGAS** and custom evaluation pipelines to measure faithfulness, hallucinations, retrieval quality, and response safety.
 - Deployed scalable LLM and RAG services using **AWS Bedrock**, **SageMaker**, **Lambda**, and **ECS/ECR**, with automated CI/CD, observability, access controls, audit logging, and rollback mechanisms.
- Hitachi Vantara** Hyderabad, India
ML Engineer Jan 2023 – Dec 2023
 - Engineered production ML inference services using **Python**, **PyTorch**, **FastAPI**, and **REST APIs**, reducing inference latency by **35%** while supporting **1M+ requests per day**.
 - Built ML pipelines using **PyTorch**, **Pandas**, **PostgreSQL**, **MLflow**, and **AWS** for automated validation, experiment tracking, model versioning, and testing, increasing production test coverage from **65% to 90%**.
 - Containerized and operated ML services using **Docker**, **Kubernetes**, **GitHub Actions**, **Prometheus**, and **Grafana**, reducing bug-detection time by **40%** while helping maintain **99.9% application uptime**.
- F13 Technologies** Remote
AWS Intern Feb 2022 – Nov 2022
 - Built AWS infrastructure for **5+ applications** using **EC2**, **S3**, **Lambda**, **RDS**, and **CloudFormation**, supporting deployments with **99.8% availability**.
 - Automated deployment and cloud-optimization workflows using **CloudFormation**, **Python**, **CloudWatch**, and **Cost Explorer**, reducing deployment time by **30%** and monthly cloud spend by **18%**.

RESEARCH WORK & PROJECTS

- San Jose State University** San Jose, CA
ML Engineer Mar 2026 – Present
 - Designed a real-time generative AI editing pipeline combining **YOLOv8**, **Stable Diffusion 3**, and **InstructPix2Pix** to detect, add, remove, and replace objects in smart-vehicle scenes through a **FastAPI/React** interface.
 - Developed adaptive generation logic that dynamically tunes diffusion steps, guidance scale, and mask expansion based on object characteristics to improve blending and visual consistency.
 - Optimized inference and output selection using **MD5 caching**, multi-seed scoring, and **OpenCV/PIL** post-processing to reduce redundant computation and improve generated-image reliability.
- Direct – AI-Powered Semantic Job Search Engine**
Skills: FastAPI, React, Supabase, Qdrant, BGE Embeddings, Cerebras LLM, Docker, GitHub Actions
 - Built a resume-aware semantic job search system using **BGE embeddings**, structured intent extraction, and **Qdrant** vector retrieval to replace traditional keyword-only matching.
 - Implemented two-stage retrieval and LLM reranking that evaluates retrieved roles and generates match scores, candidate strengths, skill gaps, and personalized fit explanations.
 - Developed the full-stack platform with resume upload, authentication, saved jobs, application tracking, automated hourly ingestion, URL hashing, and fuzzy deduplication across multiple job sources.

SKILLS

- AI/ML & GenAI:** PyTorch, TensorFlow, Scikit-learn, Transformers, Computer Vision, NLP, LLMs, RAG, GraphRAG, Prompt Engineering, LoRA, PEFT, RAGAS
- LLM & Retrieval:** LangChain, LangGraph, CrewAI, MCP, Hugging Face, BGE Embeddings, FAISS, Pinecone, Qdrant, OpenSearch, Cerebras
- Programming & Backend:** Python, Java, JavaScript, TypeScript, SQL, FastAPI, Flask, Spring Boot, REST APIs, Microservices
- Cloud & MLOps:** AWS, GCP, Azure, Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, Model Serving, Experiment Tracking, Model Monitoring
- Data & Observability:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Supabase, Pandas, NumPy, Prometheus, Grafana

EDUCATION

- San Jose State University** Jan 2024 – Dec 2025
Master of Science in Software Engineering San Jose, CA
- Jawaharlal Nehru Technological University Hyderabad (JNTUH)** Aug 2019 – May 2023
Bachelor of Technology in Information Technology Hyderabad, India